3. Problem Modeling

The problem has the following elements (Fig. 1).

- *Decision variables.* For a prescribed number N of identical turbines with rotor radius R (diameter $D = 2R$), the $n = 2N$ coordinates of the layout $\mathbf{X} = (x_1, y_1, \ldots, x_N, y_N)$.
- *Objective.* The expected farm power is maximized, i.e., the wake loss is minimized, under the Jensen wake model [38, 39] and the power curve of [3] (Sections 3.1–3.3).
- *Wind direction.* The direction θ has the probability density p_θ; it is the direction *towards* which the wind blows, counter-clockwise from east (Fig. 1a). The meteorological directions θ_{met} of Horns Rev 1 and IEA37 (wind *from*, clockwise from north) are converted by $\theta = 270° - \theta_{\text{met}}$.
- *Constraints* (Section 3.4). The turbine distances $\ell_{ij} = \|(x_i - x_j,\, y_i - y_j)\|_2$ are at least $\ell_{\min} = 4D = 8R$ [3, 40], and the farm is a disc of radius $r = 500$, 750 or 1000 m.

The spacing $4D$ is the convention of the Kusiak–Song benchmark and is kept for comparability; it lies at the dense end of practice (the installed Horns Rev 1 layout is a regular grid of about $7D$; Section 9), so our results concern a packing-constrained regime (larger spacings: Section 8.5.3).

The model is the unchanged Kusiak–Song benchmark model, and no modeling novelty is claimed; the contributions are the PSO-VNS algorithm and its controlled evaluation. It rests on the following assumptions, each stated in detail in the subsections below:

A1. All N turbines are identical, and their number is prescribed.
A2. The terrain is flat, and the farm boundary is a circle of radius r (blocks of the Horns Rev 1 and Lillgrund farms: their own boundaries).
A3. Turbines must be at least $\ell_{\min} = 4D$ apart.
A4. In each direction bin, the wind speed follows a Weibull distribution; the wind rose is discretized into N_θ direction bins.
A5. Wakes follow the Jensen top-hat model with a constant thrust coefficient C_T; a turbine is waked if its hub center lies in the wake cone, and the deficits of several wakes are combined by the root-sum-square rule.
A6. The power curve is that of the Kusiak–Song benchmark: linear between cut-in and rated speed, constant above.
A7. Only the wake loss is optimized; the electrical collector-system layout and multi-objective formulations (e.g., AEP against cost or the levelized cost of energy) are not considered.

The protocol (equal budgets, seed pairing, equal-cost controls, feasibility-aware ranking, equivalence tests) does not depend on the model and applies to such formulations as well.

3.1. Wind Model and Direction Discretization

For each direction, the wind speed s has the Weibull density

$$p_s(s; \zeta, \psi) = \frac{\zeta}{\psi}\left(\frac{s}{\psi}\right)^{\zeta-1} \exp\left[-\left(\frac{s}{\psi}\right)^{\zeta}\right]$$

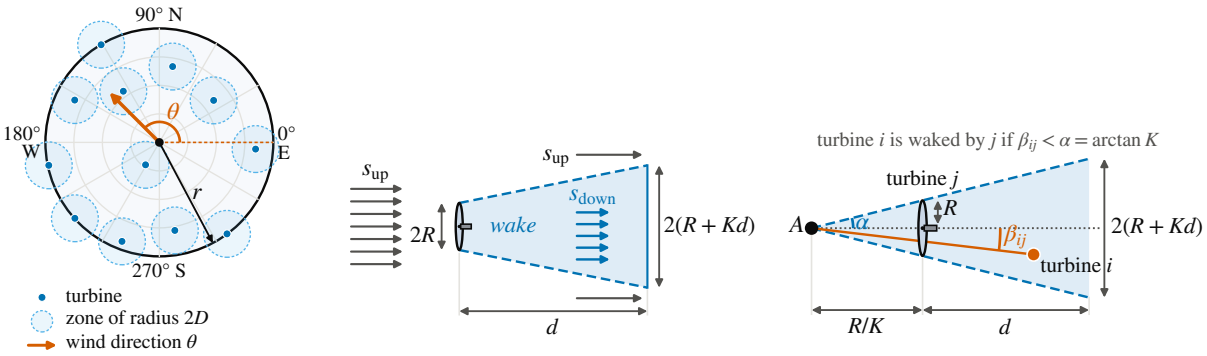


Figure 1: Model geometry. (a) Circular farm of radius r and wind direction θ; the points show, for illustration, a 12-turbine layout ($r = 750$ m); discs of radius $2D$ touch at the spacing $\ell_{\min} = 4D$. (b) Jensen top-hat wake behind a rotor of radius R: width $2(R+Kd)$ at the downstream distance d, uniform speed $s_{\text{down}} < s_{\text{up}}$. (c) Wake half cone of turbine j (K exaggerated): virtual vertex A at R/K upstream of the rotor, half-angle $\alpha = \arctan K$; turbine i is waked by j if $\beta_{ij} < \alpha$.

with shape $\zeta(\theta)$ and scale $\psi(\theta)$. The benchmark rose is piecewise constant: $N_\theta = 24$ direction bins $[15(l-1), 15l)^\circ$, $l = 1, \ldots, 24$, with probability p_l, shape $\zeta = 2$ and scale ψ_l (Table S1). The two wind data sets are:

- *Data Set I*, nearly unidirectional, with the same wind-speed distribution in all directions: $\psi = 13$ m/s in all bins, and 80% of the probability in bins 6 and 7, i.e., wind from 165° to 195°;
- *Data Set II*, a broad rose with direction-dependent wind speeds: scales from 2.6 to 10 m/s and the highest probability in bin 13.

Both are synthetic wind roses of the Kusiak–Song benchmark [3, 40], not measurements at a site; the measured wind climates in this study are the 12-sector Weibull roses of the Horns Rev 1 block (Section 9) and of a Lillgrund block (Section 10), both optimized in 5° bins, and IEA37 Case Study 1 uses 16 directions at one speed (Section 7.1).

A top-hat wake is narrow, so with 15° bins the wake of a distant pair of turbines can fall between two bin centers and not enter the objective; optimized layouts exploit these gaps (Section S1.2). Section 8.5.5 reports which conclusions survive a 1° rose.

3.2. Wake Effect Model

In the Jensen model (Fig. 1b,c), the wake of turbine j is a half cone with half-angle $\alpha = \arctan K$ (wake-spreading constant K), and its virtual vertex lies R/K upstream of the rotor. With the along-wind distance from the virtual vertex $u_{ij} = (x_i - x_j)\cos\theta + (y_i - y_j)\sin\theta + R/K$ and the lateral offset $v_{ij} = -(x_i - x_j)\sin\theta + (y_i - y_j)\cos\theta$, turbine i lies in the wake if $\beta_{ij} = \cos^{-1}\!\left(u_{ij}/\sqrt{u_{ij}^2 + v_{ij}^2}\right) < \alpha$. Its downstream distance is $d_{ij} = |u_{ij} - R/K|$, and with the thrust coefficient C_T the velocity deficit, with the free-stream speed s_{up} and the wake speed s_{down}, is

$$\Delta_{ij} = 1 - \frac{s_{\text{down}}}{s_{\text{up}}} = \frac{1 - \sqrt{1 - C_T}}{(1 + K d_{ij}/R)^2}. \tag{1}$$

The numerator is the deficit behind the rotor from actuator-disc momentum theory, and the factor $R^2/(R + K d_{ij})^2$ spreads it over the growing wake (Fig. 1b); with $C_T = 0.8$ and

$K = 0.075$, a turbine $\ell_{\min} = 308$ m directly behind another loses 21.6% of the wind speed, one 770 m ($10D$) behind 8.8%. As in [3], this top-hat wake is tested at the hub center only, so the deficit jumps at the cone edge, and it also reaches a turbine less than R/K *upstream* of j near its axis (between A and the rotor in Fig. 1c; non-physical, kept for comparability with the published results). The deficits are combined by the root-sum-square rule,

$$\Delta_i(\theta) = \left(\sum_{j \neq i,\ \beta_{ij} < \alpha} \Delta_{ij}^2 \right)^{1/2}. \tag{2}$$

As C_T is constant (which overstates the deficits above rated speed), $\Delta_i(\theta)$ depends only on the layout and the direction, not on the free-stream speed s; the local wind speed in a direction bin, $s_{\text{local}} = [1 - \Delta_i(\theta)]\, s$, is therefore a scaled Weibull variable, with the same shape $\zeta(\theta)$ and the scale $\psi_i(\theta) = \psi(\theta)[1 - \Delta_i(\theta)]$. The hub-center test makes the objective jump where a hub crosses a cone edge in an evaluated direction; hence the derivative-free local search of VNS (Section 4.3), and finite-difference gradients handicap MS-SLSQP.

The Jensen model is used for comparability with the published Kusiak–Song results; a Gaussian wake is tested in Sections 11 and 8.5.2 (reasons: Section S1.2).

3.3. Power Model and Expected Energy

The power curve of the Kusiak–Song benchmark [3] is

$$f(s) = \begin{cases} 0, & s < s_{\text{cut-in}}, \\ \lambda_1 s + \lambda_0, & s_{\text{cut-in}} \le s \le s_{\text{rated}}, \\ P_{\text{rated}}, & s > s_{\text{rated}}, \end{cases}$$

with $s_{\text{cut-in}} = 3.5$ m/s, $s_{\text{rated}} = 14$ m/s, $P_{\text{rated}} = 1500$ kW, slope $\lambda_1 = 140.86$ and intercept $\lambda_0 = -500$. We use it as published (details: Section S1.2; a cubic curve is tested in Section 8.5.1). The expected power of turbine i, whose power output $P_i = f(s_{\text{local}})$ depends on its local wind speed,

$$E(P_i) = \int_0^{360} p_\theta(\theta) \int_0^{\infty} f(s)\, p_s\Big(s; \zeta(\theta), \psi_i(\theta)\Big)\, ds\, d\theta, \tag{3}$$

is evaluated, as in [3], by a Riemann sum over 24 direction bins of 15° and 21 speed bins of 0.5 m/s between $s_{\text{cut-in}}$ and s_{rated} (Eq. (S1)). Like the earlier studies, we keep the bin width (15°) in degrees, so the benchmark objective $\sum_i E(P_i)$ is 15 times the expected farm power in kW, and the gross (wake-only) annual energy production is AEP [MWh] = (objective/15) × 8.76. The wake-free benchmark objective of one turbine, E_{ideal}, is the value of (3) with $\psi_i = \psi$; for Data Set I, $E_{\text{ideal}} = 14{,}045.74$, which corresponds to 936.38 kW (a capacity factor of 62%, far above real sites) or 8,202.7 MWh/yr.

3.4. Optimization Problem and Constraint Handling

With the wake loss $L(\mathbf{X}) = NE_{\text{ideal}} - \sum_{i=1}^{N} E(P_i)$, reported in percent of the wake-free value, $100\, L/(NE_{\text{ideal}})$, the optimization problem reads

$$\begin{aligned} \min_{\mathbf{X} \in [-r,r]^{2N}} \quad & L(\mathbf{X}) \\ \text{subject to} \quad & \ell_{ij} \ge \ell_{\min}, \qquad 1 \le i < j \le N \quad \text{(spacing)}, \\ & x_i^2 + y_i^2 \le r^2, \quad 1 \le i \le N \quad \text{(boundary)}, \end{aligned} \tag{4}$$

which is equivalent to maximizing the expected farm power and hence the AEP: a problem in $2N$ variables ($2N \leq 30$ on the benchmark) with a discontinuous objective and $N(N-1)/2$ nonconvex spacing constraints. After every position update, each coordinate is clipped to $[-r, r]$ (box repair; the square, not the circle). The constraints are handled by a penalty on the boundary and spacing violations

$$g_i^{\mathrm{b}} = x_i^2 + y_i^2 - r^2, \qquad g_{ij}^{\mathrm{s}} = \ell_{\min} - \ell_{ij};$$

all optimizers except MS-SLSQP minimize the penalized wake loss

$$\begin{aligned} F_p(\mathbf{X}) = \Big(N E_{\mathrm{ideal}} - \sum_{i=1}^{N} E(P_i)\Big) + \sum_{i:\, g_i^{\mathrm{b}}>0} \Big(1 + \mu_{\mathrm{p}} g_i^{\mathrm{b}}\Big)^2 \\ + \sum_{i<j:\, g_{ij}^{\mathrm{s}}>0} \Big(1 + \mu_{\mathrm{p}} g_{ij}^{\mathrm{s}}\Big)^2, \qquad \mu_{\mathrm{p}} = 10^{10}, \end{aligned} \tag{5}$$

whose first term is the wake loss $L(\mathbf{X})$ reported in our tables. Three properties of this treatment matter for the comparisons.

- *Feasibility first.* Any violation above about 4.6×10^{-8} (m or m^2) makes the penalty exceed any possible wake loss, so comparing F_p values is, up to smaller violations, a feasibility-first rule (feasible before infeasible, then wake loss or total penalty).
- *Scaling of the violations.* The scaling (g_i^{b} in m^2, g_{ij}^{s} in m) favors infeasible layouts with turbines slightly too close, or with a turbine clipped to the box near a tangent point $(\pm r, y_i)$, where $g_i^{\mathrm{b}} = y_i^2$.
- *Feasibility of the final layouts.* MS-SLSQP minimizes the wake loss under explicit constraints. A final layout counts as feasible if $\ell_{ij} \geq \ell_{\min}$ and $\sqrt{x_i^2 + y_i^2} \leq r$ hold within 10^{-6} m (Section 7), a looser tolerance than the penalty's. With random starts, no optimizer guarantees a feasible final layout; infeasible final layouts are therefore counted, reported and ranked below feasible ones (Section 7.3).

Section 8.5.4 tests an alternative: Deb's feasibility rules [29] on dimensionless violations, with box clipping or radial projection.

4. Preliminaries: PSO, SSA and Basic VNS

A population method keeps N_p candidate vectors $\mathbf{X}^i \in \mathbb{R}^n$ with components X_m^i and bounds $lb_m = -r$, $ub_m = r$; unless stated otherwise, one iteration costs N_p objective evaluations. This section states the methods exactly as implemented (Algorithms 1–2); Table 2 collects all settings.

4.1. Particle Swarm Optimization

We use the standard PSO, proposed by Kennedy and Eberhart [26], in the inertia-weight form [68] with the constriction coefficients of Clerc and Kennedy [27] (Algorithm 1), with a global best, run for T iterations $t = 1, \ldots, T$. Particle i has a position $\mathbf{X}^i$, a velocity $\mathbf{V}^i$ and a personal best $\mathbf{P}^i$, the swarm keeps the global best $\mathbf{G}$, and a particle moves by

$$\mathbf{V}^i \leftarrow w\,\mathbf{V}^i + c_1 \boldsymbol{\rho}_1 \circ \Big(\mathbf{P}^i - \mathbf{X}^i\Big) + c_2 \boldsymbol{\rho}_2 \circ \Big(\mathbf{G} - \mathbf{X}^i\Big), \tag{6}$$

$$\mathbf{X}^i \leftarrow \mathbf{X}^i + \mathbf{V}^i, \tag{7}$$

Algorithm 1 PSO for the WFLOP as implemented (complete procedure; $N_p(T+1)$ evaluations of F_p).

Require: site (farm radius r), number of turbines N, wind data and wake model, i.e., the penalized wake loss F_p (5); number of iterations T; random seed
Parameters: swarm size $N_p = 30$; $w = 0.7298$, $c_1 = c_2 = 1.49618$ (8)
Ensure: the global best layout $\mathbf{G}$ and its value $F_p(\mathbf{G})$
Initialization
$n \leftarrow 2N$; draw (seeded) N_p layouts $\mathbf{X}^i \in [-r, r]^n$ and evaluate $F_p(\mathbf{X}^i)$ ▷ N_p evaluations
$\mathbf{V}^i \leftarrow \mathbf{0}$, $\mathbf{P}^i \leftarrow \mathbf{X}^i$ $(i = 1, \ldots, N_p)$; $\mathbf{G} \leftarrow \arg\min_i F_p(\mathbf{P}^i)$
Main loop
for $t = 1, \ldots, T$ **do**
  for $i = 1, \ldots, N_p$ **do**
    Draw $\boldsymbol{\rho}_1, \boldsymbol{\rho}_2 \sim U(0,1)^n$; $\mathbf{V}^i \leftarrow w\,\mathbf{V}^i + c_1\boldsymbol{\rho}_1 \circ (\mathbf{P}^i - \mathbf{X}^i) + c_2\boldsymbol{\rho}_2 \circ (\mathbf{G} - \mathbf{X}^i)$ ▷ (6)
    $\mathbf{X}^i \leftarrow \mathrm{clip}(\mathbf{X}^i + \mathbf{V}^i, -r, r)$ ▷ (7) with box repair; $\mathbf{V}^i$ is kept
    Evaluate $F_p(\mathbf{X}^i)$; if $F_p(\mathbf{X}^i) < F_p(\mathbf{P}^i)$, then $\mathbf{P}^i \leftarrow \mathbf{X}^i$; if $F_p(\mathbf{X}^i) < F_p(\mathbf{G})$, then $\mathbf{G} \leftarrow \mathbf{X}^i$ ▷ asynchronous update
  end for
end for
Termination
Stop after T iterations, i.e., $N_p(T+1)$ evaluations; **return G**

where $\boldsymbol{\rho}_1, \boldsymbol{\rho}_2$ are vectors of independent $U(0,1)$ numbers, drawn anew for every move, and $\circ$ is the componentwise product. The constriction PSO of Clerc and Kennedy [27] with their standard choice $\varphi_1 = \varphi_2 = 2.05$ (constriction factor $\kappa = 0.7298$) is identical to (6) with

$$w = 0.7298, \qquad c_1 = c_2 = \kappa\varphi_1 = 1.49618. \tag{8}$$

PSO denotes (6) and (7) with these coefficients, fixed a priori from the literature. It is run as implemented: velocities start at zero and are not clamped (no bound $|V_m^i| \le V_{\max}$), a coordinate clipped to the square (Section 3.4) keeps its velocity, and $\mathbf{P}^i$, $\mathbf{G}$ are updated after each evaluation (asynchronous global best).

Stability. If $\mathbf{P}^i$ and $\mathbf{G}$ stay fixed (stagnation) and no bound is active, every coordinate follows a random linear recursion; for $c_1 = c_2$, its mean and variance converge if and only if $|w| < 1$ and $c_1 + c_2 < 24(1 - w^2)/(7 - 5w)$, the order-2 condition of Poli [69, 71]. These known results, restated for the implemented update with their proof in Proposition S1, are used only to check the setting, not claimed as new theory. The constriction setting satisfies the condition ($c_1 + c_2 = 2.99 < 3.35$; Fig. S11; Proposition S2): in the limit, a coordinate with $P_m^i \neq G_m$ is sampled with mean $(P_m^i + G_m)/2$ and standard deviation $1.04\,|P_m^i - G_m|$, so the swarm keeps sampling around $\mathbf{G}$ unless $\mathbf{P}^i = \mathbf{G}$. Stability does not say how good $\mathbf{G}$ is, and clipping at the bounds breaks the second assumption (Lemma S3). The scale factor $F = 0.5$ of differential evolution (DE; Section 7.2) exceeds Zaharie's critical value (0.14 at CR = 0.9) [74], so its weak results are not due to premature variance loss.

4.2. Salp Swarm Algorithm and Its Laplacian Variant

The SSA was proposed by Mirjalili et al. [8] (pseudocode: Algorithm S1). Its population of N_p salps, sorted by F_p, consists of two halves:

- *Food source.* The best solution found so far, i.e., the candidate with the best fitness (lowest F_p), is the food source $\mathbf{H}$; after every iteration, the best salp of the sorted population, the first leader, replaces $\mathbf{H}$ if it is better.

- *Leaders.* The better half, $i \le N_p/2$, updates each coordinate around the food source by
$$X_m^i = H_m \pm \xi_1\Big((ub_m - lb_m)\xi_2 + lb_m\Big), \tag{9}$$
with either sign with probability $1/2$, $\xi_2 \sim U(0,1)$ and $\xi_1 = 2\exp[-(4t/T)^2]$ decreasing with the iteration t out of T: a leader samples $H_m + U(-\xi_1 r, \xi_1 r)$ in every coordinate, with $\xi_1 \le 0.037$ for $t \ge T/2$.

- *Followers.* A follower $i > N_p/2$ moves to the midpoint of its position and the *new* position of the preceding salp.

The leaders thus perturb $\mathbf{H}$ uniformly with a shrinking radius, and the followers only average [33]. The SSA has known drawbacks, such as premature convergence and an imbalance between exploration and exploitation [28, 33]. Solanki and Deep [28] addressed them by proposing a new variant, the Laplacian SSA (LX-SSA). LX-SSA gives each follower a second candidate $Y_m^i = X_m^i + \gamma(H_m - X_m^i)$ with one Laplace step [75] for all coordinates,
$$\gamma = \phi - \chi \operatorname{sgn}\Big(z - \tfrac{1}{2}\Big) \ln\Big(1 - 2|z - \tfrac{1}{2}|\Big), \quad z \sim U(0,1), \tag{10}$$
with the published, untuned values $\phi = 0$ (centered on $\mathbf{X}^i$) and $\chi = 1$ (mean step length equal to the distance to $\mathbf{H}$), and keeps the better candidate. As published, it evaluates both candidates of each follower and then re-evaluates the kept ones with the new population, so an iteration costs $2N_p$ evaluations (SSA: N_p) and LX-SSA runs half as many iterations as SSA (Table 2). Comparing the two tests LX-SSA as published, not the Laplace step alone (for the step itself, see the ablation in Section 12.3); both use the code of the published LX-SSA study [28].

4.3. Basic Variable Neighborhood Search

The basic VNS [23, 24], in the continuous form of [73] (Algorithm 2), improves one incumbent $\mathbf{X}$ with $k_{\max} = 5$ ℓ_∞ shell neighborhoods $\mathcal{N}_k(\mathbf{X}) = \{\mathbf{X}' : \delta_{k-1} < \|\mathbf{X}' - \mathbf{X}\|_\infty \le \delta_k\}$, $\delta_k = 0.1\,k\,r$ ($\delta_0 = 0$), of the whole layout: shaking draws a uniform point of $\mathcal{N}_k(\mathbf{X})$, clipped to the box, and a best-improvement local search improves it. As the top-hat wake makes F_p discontinuous, the local search is a compass search: it evaluates all $2n$ moves $\pm h\,\mathbf{e}_m$ ($\mathbf{e}_m$ the mth unit vector; 60 evaluations for $N = 15$), moves to the best one if it strictly improves, and halves h (from $h_0 = 0.05r$) when none does, until $h < h_{\min} = 10^{-3}r$. A completed descent ends at a point that no move $\pm\bar{h}\,\mathbf{e}_m$, $\bar{h} = h_0/2^5 = 0.0015625\,r$, improves (Proposition S10), a coordinate-wise optimality at one step length, not a local minimum.

5. The Proposed PSO-VNS

This section motivates PSO-VNS, states the algorithm and gives its properties. PSO-VNS combines the PSO of Algorithm 1, in the constriction setting (8), with the basic VNS of Algorithm 2 in two consecutive phases.

5.1. Motivation

Three properties of the continuous WFLOP shape the design.

Algorithm 2 Basic VNS for the WFLOP as implemented (complete procedure; every call of F_p counts against the budget B).

Require: site (farm radius r), number of turbines N, wind data and wake model, i.e., the penalized wake loss F_p (5); budget B; random seed; an incumbent layout $\mathbf{X}$ (VNS alone: none; Phase 2 of a hybrid: the layout passed by Phase 1)
Parameters: $k_{\max} = 5$, $\delta_k = 0.1\,k\,r$ ($\delta_0 = 0$); $h_0 = 0.05\,r$, $h_{\min} = 10^{-3}r$
Ensure: the best layout $\mathbf{X}^\star$ evaluated
Initialization
$n \leftarrow 2N$; **if** no incumbent is given **then** draw (seeded) $N_p = 30$ layouts in $[-r, r]^n$, evaluate them and let $\mathbf{X}$ be the best ▷ VNS alone
$\mathbf{X} \leftarrow \textsc{LocalSearch}(\mathbf{X})$; $k \leftarrow 1$
Main loop
while fewer than B evaluations have been used **do**
  Draw $\mathbf{d} \sim U[-\delta_k, \delta_k]^n$ until $\|\mathbf{d}\|_\infty > \delta_{k-1}$ ▷ shaking in $\mathcal{N}_k(\mathbf{X})$
  $\mathbf{X}' \leftarrow \mathrm{clip}(\mathbf{X} + \mathbf{d}, -r, r)$; evaluate $F_p(\mathbf{X}')$
  $\mathbf{X}'' \leftarrow \textsc{LocalSearch}(\mathbf{X}')$
  if $F_p(\mathbf{X}'') < F_p(\mathbf{X})$ **then** $\mathbf{X} \leftarrow \mathbf{X}''$, $k \leftarrow 1$ ▷ move
  else $k \leftarrow k \bmod k_{\max} + 1$ ▷ next, larger neighborhood
  end if
end while
Termination
Stop as soon as B evaluations have been used (also inside LocalSearch); **return** $\mathbf{X}^\star$, the best layout evaluated

function $\textsc{LocalSearch}(\mathbf{y})$ ▷ compass search, best improvement
  $h \leftarrow h_0$
  while $h \geq h_{\min}$ **do**
    Evaluate the $2n$ trial layouts $\mathrm{clip}(\mathbf{y} \pm h\,\mathbf{e}_m, -r, r)$, $m = 1, \ldots, n$ ($\mathbf{e}_m$: mth unit vector)
    if the best trial layout $\mathbf{y}'$ has $F_p(\mathbf{y}') < F_p(\mathbf{y})$ **then** $\mathbf{y} \leftarrow \mathbf{y}'$
    else $h \leftarrow h/2$
    end if
  end while
  return $\mathbf{y}$
end function

- *Feasible layouts are hard to reach by sampling.* The feasible region is a small, non-convex part of the bounding box; for the largest benchmark cases, fewer than one uniform layout in 10^6 is feasible (Monte Carlo estimate for $N = 12$ and 15; Proposition S5 and Table S48). A method that refines a single layout therefore needs a good, preferably feasible, start, and a population-based global search that moves whole layouts towards the feasible region is a natural source of such starts. PSO is widely used for the WFLOP [6, 7], has few parameters, and its stability under stagnation is understood [69–71], so that a setting can be checked before it is used (Section 4.1).

- *PSO is a weak finisher.* Under a stable setting and while the best positions stay fixed, the expected position of a particle converges to a weighted average of its personal best and the global best [76, 77], a point that need not be a local optimum [71]; late in a run, refinement thus relies on random sampling around the global best. The last improvements of a layout, however, are small moves of individual turbine

coordinates, which a coordinate-wise local search makes systematically.

- *The objective is discontinuous.* With the top-hat wake, F_p jumps where a turbine enters or leaves a wake cone (Section 3), so gradient-based refinement is unreliable and a derivative-free local search is needed. The basic VNS [23, 24], in its continuous form [73], fits this need: its local search, a compass search, compares objective values only; its shaking in nested neighborhoods of the whole layout gives a systematic way to leave a local optimum (at the 6,030-evaluation budget studied here it rarely acts; Section 12.1); and its few parameters can be kept at their defaults.

The need for a new variant. PSO and VNS have been combined before, with VNS as the local search applied within the PSO loop [10–12] (Section 2.5). Such embedded designs spend local-search evaluations in every iteration, which makes the cost of each component hard to separate, and, to our knowledge, they have not been applied to the WFLOP. PSO-VNS instead runs the two methods in sequence: PSO for a share ω of the budget, then VNS from the best layout found. This design

(i) uses the swarm where it is strong, in reaching the feasible region from random starts, and the local search where the swarm is weak, in refining one layout;
(ii) gives both phases fixed, exactly counted budgets;
(iii) adds a single parameter, ω, to those of PSO and VNS; and
(iv) keeps Phase 1 a plain PSO, so that the contribution of the local-search phase can be measured directly against PSO with the same seeds (Section 5.3).

5.2. Algorithm

PSO-VNS (Algorithm 3) is a new two-phase variant of the PSO–VNS hybrid. *Phase 1* runs the PSO of Algorithm 1 from the seeded initial population for $T_1 = \text{round}((\omega B - N_p)/N_p)$ iterations, that is, $N_p(T_1 + 1)$ evaluations, and passes its global best layout $\mathbf{G}$ to Phase 2, feasible or not; the penalty then guides the local search towards feasibility. *Phase 2* applies the basic VNS (Algorithm 2) to $\mathbf{G}$: a compass-search descent, then, until the budget B is used, shaking in the neighborhood $\mathcal{N}_k$ followed by a descent, accepting an improvement and returning to $k = 1$, or else moving to the next, larger neighborhood. The best layout evaluated is returned.

With $\omega = 0.5$ and $B = 6{,}030$, PSO runs for $T_1 = 100$ iterations (99.5 rounded half to even) and VNS receives 3,000 evaluations (Table 2). None of these values was tuned; Section 12.4 studies ω. At this budget, PSO-VNS is essentially PSO plus one compass-search descent, and the neighborhoods matter only for small N (Section 12.1).

5.3. Properties and Computational Cost

Exact prefix. As the PSO coefficients do not depend on the iteration, Phase 1 reproduces exactly, for a given seed, the first $N_p(T_1 + 1)$ evaluations of a PSO run (Proposition S4); comparing PSO-VNS with PSO thus isolates the effect of Phase 2. This does not hold for SSA-VNS and LX-SSA-VNS, whose ξ_1 depends on t/T_1 (Section 6).

Cost. A run uses exactly B evaluations of F_p, including the initial population, and no gradients; the overhead of both phases is $\mathcal{O}(n)$ arithmetic per evaluation and $\mathcal{O}(N_p n)$ memory, small against a wake-model evaluation, so the run time is governed by B (Section 7.5).

Constraints. Neither phase enforces the constraints: both minimize the penalized loss F_p (5) over the box $[-r, r]^{2N}$, with clipping, and the feasibility of the returned layout is checked and reported for every run (Section 7).

Algorithm 3 PSO-VNS for the WFLOP (complete procedure; every call of F_p counts against the budget B).

Require: site (farm radius r), number of turbines N, wind data and wake model, i.e., the penalized wake loss F_p (5); budget B; random seed
Parameters: population $N_p = 30$; $w = 0.7298$, $c_1 = c_2 = 1.49618$ (8); split $\omega = 0.5$; $k_{\max} = 5$, $\delta_k = 0.1\,k\,r$ ($\delta_0 = 0$); $h_0 = 0.05\,r$, $h_{\min} = 10^{-3}r$
Ensure: the best layout $\mathbf{X}^\star$ evaluated, with its wake loss and feasibility
Initialization
$n \leftarrow 2N$; $T_1 \leftarrow \text{round}((\omega B - N_p)/N_p)$ ▷ $T_1 = 100$ for $B = 6{,}030$
Draw (seeded) N_p layouts $\mathbf{X}^i \in [-r, r]^n$ and evaluate $F_p(\mathbf{X}^i)$ ▷ N_p evaluations
$\mathbf{V}^i \leftarrow \mathbf{0}$, $\mathbf{P}^i \leftarrow \mathbf{X}^i$ $(i = 1, \ldots, N_p)$; $\mathbf{G} \leftarrow \arg\min_i F_p(\mathbf{P}^i)$
Phase 1: PSO (global search towards the feasible region)
for $t = 1, \ldots, T_1$ **do**
  for $i = 1, \ldots, N_p$ **do**
    Draw $\boldsymbol{\rho}_1, \boldsymbol{\rho}_2 \sim U(0,1)^n$; $\mathbf{V}^i \leftarrow w\,\mathbf{V}^i + c_1\boldsymbol{\rho}_1 \circ (\mathbf{P}^i - \mathbf{X}^i) + c_2\boldsymbol{\rho}_2 \circ (\mathbf{G} - \mathbf{X}^i)$
    $\mathbf{X}^i \leftarrow \text{clip}(\mathbf{X}^i + \mathbf{V}^i, -r, r)$ ▷ box repair; $\mathbf{V}^i$ is kept
    Evaluate $F_p(\mathbf{X}^i)$; if $F_p(\mathbf{X}^i) < F_p(\mathbf{P}^i)$, then $\mathbf{P}^i \leftarrow \mathbf{X}^i$; if $F_p(\mathbf{X}^i) < F_p(\mathbf{G})$, then $\mathbf{G} \leftarrow \mathbf{X}^i$
  end for
end for ▷ $N_p(T_1 + 1)$ evaluations used
Phase 2: basic VNS (local refinement of $\mathbf{G}$)
$\mathbf{X} \leftarrow$ LocalSearch($\mathbf{G}$); $k \leftarrow 1$
while fewer than B evaluations have been used **do**
  Draw $\mathbf{d} \sim U[-\delta_k, \delta_k]^n$ until $\|\mathbf{d}\|_\infty > \delta_{k-1}$ ▷ shaking in $\mathcal{N}_k(\mathbf{X})$
  $\mathbf{X}' \leftarrow \text{clip}(\mathbf{X} + \mathbf{d}, -r, r)$; evaluate $F_p(\mathbf{X}')$
  $\mathbf{X}'' \leftarrow$ LocalSearch($\mathbf{X}'$)
  if $F_p(\mathbf{X}'') < F_p(\mathbf{X})$ **then** $\mathbf{X} \leftarrow \mathbf{X}''$, $k \leftarrow 1$ ▷ move
  else $k \leftarrow k \bmod k_{\max} + 1$ ▷ next, larger neighborhood
  end if
end while
Termination
Stop as soon as B evaluations have been used (also inside LocalSearch); **return** $\mathbf{X}^\star$, the best layout evaluated in either phase

function LocalSearch($\mathbf{y}$) ▷ compass search, best improvement
  $h \leftarrow h_0$
  while $h \geq h_{\min}$ **do**
    Evaluate the $2n$ trial layouts $\text{clip}(\mathbf{y} \pm h\,\mathbf{e}_m, -r, r)$, $m = 1, \ldots, n$ ($\mathbf{e}_m$: mth unit vector)
    if the best trial layout $\mathbf{y}'$ has $F_p(\mathbf{y}') < F_p(\mathbf{y})$ **then** $\mathbf{y} \leftarrow \mathbf{y}'$
    else $h \leftarrow h/2$
    end if
  end while
  return $\mathbf{y}$
end function

6. Comparison Hybrids and Random-Sampling Controls

6.1. Two-Phase Template

The other hybrids and the random-sampling controls follow the sequential (relay) template of PSO-VNS (Section 5), so that they differ from it only in Phase 1. For a budget of B evaluations and a split $\omega \in (0, 1)$, Phase 1 starts from the seeded initial

population, runs a population method for about ωB evaluations and passes its best layout to Phase 2, the basic VNS (Algorithm 2), which applies the local search to it and then iterates until B evaluations are used. Only Phase 1 differs:

- PSO in *PSO-VNS*;
- SSA or LX-SSA in *SSA-VNS* and *LX-SSA-VNS* (with ξ_1 compressed to the Phase-1 iterations); and,
- in the controls, round(ωB) layouts: the initial population, then layouts drawn uniformly in the farm disc (*RSD-VNS*) or in its bounding square (*RS-VNS*).

VNS alone is the limiting case in which Phase 1 is the initial population. Phase 1 runs for

$$T_1 = \text{round}((\omega B - N_p)/c_{\text{it}})$$

iterations of c_{it} evaluations ($c_{\text{it}} = N_p$ for PSO and SSA, $2N_p$ for LX-SSA).

The budget is not shared between methods: every run has its own B evaluations, split in the same way between the two phases ($\omega = 0.5$; 3,030 Phase-1 evaluations for the swarms, 3,015 for the controls). The hybrids and controls thus differ only in how Phase 1 spends its half, and a swarm phase is useful only if it gives VNS a better start than random sampling; the ablation (Section 12) tests this against RSD-VNS, with RS-VNS as a weak minimum bar.

Random layouts are rarely feasible: for the largest cases ($N = 12, 15$), a layout drawn uniformly in the bounding square is feasible with probability below 10^{-6} (Monte Carlo; Proposition S5 and Table S48), so the 3,015 samples of RS-VNS contain about 10^{-3} or fewer feasible layouts in expectation, and RS-VNS hands VNS the least-violating sample. The spacing dominates: 730 RS-VNS runs have no feasible sample, and disc sampling gives one in only 242 of them.

Table 2 lists the settings of all methods, the published defaults [8, 27, 28, 73, 78]; MS-SLSQP (SciPy's SLSQP [79]) minimizes L under the scaled constraints of (4) (Section S1.6). At 30,030 and 120,030 evaluations (Section 8.4.1), T and T_1 follow the same rules.

(Section 7 and later: next review steps)

Table 2: Settings and evaluation budgets of all methods at the benchmark budget $B = 6{,}030$ evaluations of F_p (including the initial population and, for MS-SLSQP, the finite-difference calls); $N_p = 30$; box repair by clipping to $[-r, r]$. Every run of every method uses exactly B evaluations; the numbers of iterations differ because one iteration costs $N_p = 30$ evaluations ($2N_p = 60$ for LX-SSA), and a hybrid gives each of its two phases about half of B. A population method uses $N_p(T + 1)$ evaluations, LX-SSA $N_p(2T + 1)$.

Method	Iterations	Evaluations (= 6,030)	Coefficients and operators
PSO	$T = 200$	$30 + 200 \times 30$	Algorithm 1; $w = 0.7298$, $c_1 = c_2 = 1.49618$; $\mathbf{V}_0^i = \mathbf{0}$, no $V_{\max}$; clipped coordinate keeps its velocity; asynchronous $\mathbf{P}^i$, $\mathbf{G}$
SSA	$T = 200$	$30 + 200 \times 30$	Algorithm S1; 15 leaders, 15 followers; $\xi_1 = 2\exp[-(4t/T)^2]$
LX-SSA	$T = 100$	$30 + 100 \times 60$	as SSA, plus a Laplace candidate per follower, $\phi = 0$, $\chi = 1$ (10); both candidates evaluated
DE/rand/1/bin	$T = 200$	$30 + 200 \times 30$	$F = 0.5$, CR $= 0.9$; mutant clipped before crossover; asynchronous one-to-one replacement
GA	$T = 200$	$30 + 200 \times 30$	binary tournament; SBX [80] ($\eta_c = 15$, $p_c = 0.9$); polynomial mutation [81] ($\eta_m = 20$, $p_m = 1/n$); clipping; elitism
VNS	–	$30 + 6{,}000$	Algorithm 2 from the best initial layout; $k_{\max} = 5$, $\delta_k = 0.1kr$, $h_0 = 0.05r$, $h_{\min} = 10^{-3}r$
MS-SLSQP	≤ 200 per start	starts until B is used	SLSQP, ftol $= 10^{-9}$, forward differences (step 1 m); starts: initial layouts by increasing F_p, then random layouts
PSO-VNS, SSA-VNS	$T_1 = 100$	$(30 + 100 \times 30) + 3{,}000$	$\omega = 0.5$; Phase 1 PSO / SSA as above (ξ_1 over T_1), Phase 2 VNS (Algorithm 2)
LX-SSA-VNS	$T_1 = 50$	$(30 + 50 \times 60) + 3{,}000$	$\omega = 0.5$; Phase 1 LX-SSA as above (ξ_1 over T_1), Phase 2 VNS
RS-VNS, RSD-VNS	–	$3{,}015 + 3{,}015$	$\omega = 0.5$; initial population, then uniform layouts in the square / disc; Phase 2 VNS

(Table 2 belongs to Section 6; the rest of this page is Section 7: next review step)